

Effect of Yogic Practice on Selected Psychological Variables among Athletes

Dr. P. Gopinathan*

ABSTRACT

The purpose of the present study was to find out the effect of yogic practices on selected psychological variables among athletes. For this purpose, 30 inter-collegiate men athletes were selected as subjects from the Arts and Science College; and they were divided randomly into two equal groups with 15 each as experimental and control group. Both the group undergone usual Athletics training, in the morning and evening, three to four hours; and in addition experimental group underwent Yogic practice morning 60-minutes in a schedule of weekly three alternative days; and the control group was not given any special training other than their routine Athletics training. The training period for this study was 6 weeks in a schedule of three days in a week. The pre and post test were conducted prior and after the training programme on the selected psychological variables of stress, anxiety and self-confidence. The collected data were statistically analyzed by using ANCOVA, to find out the significant difference between the groups, if any. The significant level was fixed at 0.05 levels. It was concluded from the result of the study that the experimental group that had done the yogic practice had significant impact on the selected psychological variables of stress, anxiety and self-confidence, among inter-collegiate athletes. Further, the mean value indicated that the experimental group considerably reduced the stress and anxiety and increased self-confidence, than the control group.

Key words

Yoga, stress, anxiety and self-confidence

INTRODUCTION

Yoga is perhaps the only form of activity which massages all the internal glands and organs of the body, in a thorough manner. Yoga acts in a healthy manner on the various body parts. Yoga, originated in India thousands years ago, is a method of learning that aims to attain the unity of mind, body, and spirit through three main Yoga structures: exercise, breathing, and meditation (Gilbert, 1999). The physical postures and exercises (Asanas), breathing

techniques (pranayama), deep relaxation, and meditation practices are cultivating proper awareness and ultimately more profound states of consciousness.

Yoga (Yoga Psychology) offers practices to enhance mental health and alleviate the symptoms of psychological illness or mental illness, using mind-body-lifestyle interventions to cultivate a healthy, productive, flourishing life. According to the British Psychological Society, yoga involves concentration on the breath and

* Assistant Professor, Department of Sports Psychology and Sociology, Tamil Nadu Physical Education and Sports University, Melakottaiyur Post, Chennai-600 127

body, which makes it a great way to soothe a person's mind and relieve worries. By helping discharge tension and stress, yoga poses and breathing exercises keep a person free from such negative elements. As a result, a person who does yoga is better able to achieve the peak of psychological health.

Yoga provides the path to achieve greater perfection of the body, life and mind. Yoga physically creates a toned, flexible, and strong body and physiologically improves respiration, energy, vitality and helps to maintain a balanced metabolism, promotes cardio and circulatory health, relieves pain and also improves athletic performance. Anxiety is a mood state characterized by marked negative affect and somatic symptoms of tension in which a person apprehensively anticipates future danger or misfortune (Barlow, 1988). Hans Selye (1983) defined stress as the "nonspecific response of the body to any demand made upon it." Vealey defined sports confidence as "the belief or degree of certainty individual possess about their ability to be successful in sport." Self confidence can be described as a belief in one's self and one's ability to succeed.

There are plenty of research studies conducted in yoga practices in relation to physiological variables and almost all the studies given a very strong input in improving the psychological qualities. Rachel E. Maddux et al (2018) found that it has positive effects on a psychophysiological level that leads to decreased levels of stress in college student. Nityananthan and Kalpana (2014) found out that yoga improved overall health conditions of the middle-aged men, assessed through psychological variables self-confidence with significant

improvement and reduction in stress. Boggiti Rajesh Jayaratnam (2013) found out twelve weeks yogic exercises significantly altered anxiety, aggression, self-confidence of the inter university Hockey players. Kate Hays et al (2009) examined the role of confidence in relation to the cognitive, affective, and behavioural responses, and they found out that high sport confidence facilitated performance through its positive effect on athletes' thoughts, feelings, and behaviours. Roth and Creaser (1997) conducted an 8-week meditation based on Stress Reduction and Relaxation Programme, and they found a considerable reduction in stress due to meditation. Susan Lark (1993) said that practicing Yoga will "provide effective relief from anxiety and stress. Yogic practice has a significant reduction in total anxiety (Kocher, 1972).

Objectives of the Study

The core aim of the present study was to find out the effect of yogic practices on selected psychological variables of stress, anxiety and self-confidence among inter-collegiate athletes.

METHODOLOGY

The selection of subjects, variables, training procedure, description of testing procedure and statistical techniques are explained below.

Subject

Thirty inter-collegiate men athletes were selected as subjects for the study, and they were randomly divided into two groups equally with fifteen each as experimental and control group. The age groups of the subjects were eighteen to twenty-four years.

Training Procedure

The experimental and control groups

practiced athletic training morning and evening three to four hours; and apart from Athletics training, experimental group underwent one hour yoga practice, in a schedule of weekly three alternative days, morning one hour, for a period of six-week experimental period. The one hour yoga training included twelve yogasanas and meditation, three pranayamas. The Yogasanas are Padmasana, Sarvangasana, Tadasana, Halasana, Bhujangasana, Matsyasana, Chakrasana, Dhanurasana, Vazerasana, Paschimottanasana, Sirashasana and Savasana, Pranayamas are Bhastrika, Kapalabati and Anulom Vilom; and Transcendental Meditation.

Testing Procedure and Description of Psychological Tools used

The pre and post test were conducted prior and after the training programme, on the selected psychological variables of stress, sports competition anxiety and self-confidence. All the tests were carried out with a standardized procedure. Stress was assessed through Psychological Stress Scale designed by Everly and Girdano. This questionnaire consists of 14 statements with four options. The score ranges from 0 to 42 and lesser the score lesser the psychological stress. Anxiety was measured through Rainer Martens (1997) Sports Competition

Anxiety Test. This questionnaire consists 15 statements with score ranges from 11 to 33. The score less than 17 indicating low level anxiety; a score between 17 and 24 indicating the average level of anxiety; and score more than 24 indicating a high level of anxiety. Self-confidence was measured through a Hardy Nelson self-confidence Questionnaire. This questionnaire consists of 56 statements with yes or no option and total value was 0 to 56. Higher the score higher the self confidence

Statistical Procedure

The analysis of covariance (ANCOVA) was used as a statistical tool to determine the significant difference on the data of pre and post mean obtained for stress, anxiety and self-confidence among inter-collegiate men athletes, between control and experimental group. The level of significance was fixed at 0.05 level of confidence.

RESULTS & DISCUSSION

The analysis of covariance on the data obtained on stress, anxiety and self-confidence of pre and post tests are tabulated and presented in the Tables 1, 2 and 3.

Results on Stress

The analysis of covariance on the data obtained for stress of pre and post test on experimental and control group are presented in Table 1

Table-1: Computation of Analysis of Covariance on Stress

TEST	Exp. Group	Con. Group	sv	Sum of Squares	df	Mean Square	F ratio
Pre test	28.46	29.8	B	13.333	1	13.333	-0.062
			W	-5938.36	28	-212.084	
Post test	24.6	28.73	B	128.133	1	128.133	7.31*
			W	490.533	28	17.519	
Adjusted Mean	24.57	28.75	B	131.552	1	131.551	7.11*
			W	498.908	27	18.478	

*Significant at 0.05 level of confidence

It was observed from the Table 1 that there was no significant difference in the pre test ($F=0.062 < 4.20$). The significant differences were observed in post test ($F=7.31 > 4.20$) for df 1 and 28 at 0.05 level of confidence and adjusted post test ($F=7.11 > 4.21$) for df 1 and 27 at 0.05 level of confidence. There was a significant difference in stress and mean score

indicated that the experimental group reduced the stress level than control group due to six weeks yoga practice.

Results on Sports Competition Anxiety

The analysis of covariance on the data obtained for sports competition anxiety of pre and post test on experimental and control group are presented in Table 2.

Table 2: Computation of Analysis of Covariance on Sports Competition Anxiety

TEST	Exp. Group	Con. Group	sv	Sum of Squares	df	Mean Square	F ratio
Pre test	19.66	19.13	B	2.133	1	2.133	0.174
			W	343.06	28	12.252	
Post test	17.13	20.2	B	70.533	1	70.533	-8.37*
			W	-235.867	28	-8.423	
Adjusted Mean	16.92	20.40	B	90.540	1	90.540	-5.46*
			W	-447.209	27	-16.563	

*Significant at 0.05 level of confidence

It was observed from the Table 2 that there was no significant difference in the pre test ($F=0.174 < 4.20$). The significant differences were observed in post test ($F=8.37 > 4.20$) for df 1 and 28 at 0.05 level of confidence and adjusted post test ($F=5.46 > 4.21$) for df 1 and 27 at 0.05 level of confidence. There was a significant difference in Sports Competition Anxiety

and mean score indicated that the experimental group reduced the level of anxiety than control group due to six weeks yoga practice

Results on Self-confidence

The analysis of covariance on the data obtained for self-confidence of pre and post test on experimental and control group are presented in Table 3

Table-3: Computation of Analysis of Covariance on Self-confidence

TEST	Exp. Group	Con. Group	sv	Sum of Squares	df	Mean Square	F ratio
Pre test	34.26	33.93	B	0.833	1	0.833	0.134
			W	173.866	28	6.209	
Post test	40.06	34.46	B	235.2	1	235.2	16.03*
			W	410.666	28	14.666	
Adjusted Mean	39.88	34.64	B	205.336	1	205.336	25.98*
			W	213.395	27	7.903	

*Significant at 0.05 level of confidence

It was observed from the Table 3 that there was no significant difference in the

pretest ($F=0.134 < 4.20$). The significant differences were observed in post test

($F=16.03 > 4.20$) for df 1 and 28 at 0.05 level of confidence and adjusted post test ($F=25.98 > 4.21$) for df 1 and 27 at 0.05 level of confidence. There was a significant difference in self-confidence and mean score indicated that the experimental group increased the level of self-confidence than control group, due to six weeks yoga practice

CONCLUSION

Based on the result of the study, the following conclusions were drawn:

1. Yoga practice is benevolent to improve

the psychological qualities of stress, anxiety and self-confidence among inter-collegiate men athletes. The stress and sports competition anxiety were considerably reduced and self-confidence was improved among experimental group than the control group.

2. Result of the study proved that there was a significant difference in stress, anxiety and self-confidence among inter-collegiate athletes, due to six weeks yogasana practice.

REFERENCES

- Barlow, D. H. (1988).** Anxiety and its disorders: The nature and treatment of anxiety and panic. New York: Guilford
- Boggiti Rajesh Jayaratnam (2013).** Field Hockey better psychological performance through the yogic training. International Journal of Scientific & Engineering Research, 4 (2), 1-4.
- Gilbert, C (1999).** Yoga and breathing. Journal of Bodywork Movement Therapy, 3, 44-54
- Roth, B& Creaser, T. (1997).** Mindfulness meditation-based stress reduction: experience with a bilingual inner-city program." The Nurse Practitioner, 22(3),150-157
- Kocher,H.C., &Pratap, V. (1972).** Anxiety level and Yogic Practices. Yoga Mimamsa, 15(1),11-15
- Lark Susan, M. (1993).** "Anxiety and Stress", The Women's Health Series, Westchester Publishing Co., Los Altos, Ca, P.201
- Martens, R. (1977).** Sports Competition Anxiety Test, Champaign, IL: Human Kinetic
- Nelson, D & Hardy, L. (2001).** The development of an empirically validated tool for measuring psychological skill in sport. Journal of Sports Sciences, 8, 71
- Nityananthan, V&Kalpana, B (2014).** Management in health, impact of yoga on stress and self-confidence among the middleaged men, 18 (4), 36-39
- Rachel E. Maddux, Daiva Daukantaitė & Una Tellhed (2018).** The effects of yoga on stress and psychological health among employees: an 8- and 16-week intervention study. Journal of Anxiety, Stress, & Coping, 31(2), 121-134
- Richard H.Cox (2002).** Sports Psychology Concepts and Applications, 5th Edition, The McGraw-Hill Companies, Inc. USA, p.193
- Kate Hays, Owen Thomas, Ian Maynard & Mark Bawden (2009).** The role of confidence in world-class sport performance. Journal of Sports Sciences, 27(11), 1185-1199.
<https://www.vocabulary.com/dictionary/self-confident>
<https://www.yogapsychology.com.au/>
<https://examinedexistence.com/the-five-mental-and-psychological-benefits-of-yoga/>
<https://www.psychologytoday.com/intl/basics/confidence>